SPOT-The-Difference: Evaluation of Skeleton-Posterior-Guided Latent-Aware Causal Discovery on PetShop

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***Abstract*—SPOT speeds up differentiable causal discovery with latent confounders by replacing ABIC's dense search with a learned skeleton posterior that masks and prunes candidate edges. We evaluate it independently against its ABIC backbone on simulated ancestral ADMGs ($d = 25$) and on the PetShop microservice benchmark. On simulated graphs SPOT improves skeleton F1 by 0.067 (95% CI 0.044 to 0.090) and lowers PAG SHD. On PetShop the accuracy gain is small (+0.029 skeleton F1, CI 0.014 to 0.045, pooled over 20 runs, not detectable per scenario), but the optimizer is about eight times faster, and at an equal 60 s budget on a quiet machine SPOT converged in 11 of 12 runs against 1 of 12 for ABIC. Most of the speed-up comes from the hard floor on edge probabilities. The posterior is calibrated on simulated graphs but over-confident on PetShop, and an oracle posterior would lift PetShop skeleton F1 from 0.54 to 0.81, so posterior quality is the main limit. A fitted ADMG predicts held-out latency better than no structure, but is not shown to match a saturated Gaussian. Classical FCI/RFCI are more accurate than both score-based methods on our synthetic graphs, and learned graphs did not improve root-cause ranking. With its default start, ABIC diverges at $d = 100$; from $d = 25$ to 100 SPOT keeps its skeleton-F1 advantage over a scaled-start ABIC on every seed we ran (3–4 seeds, descriptive). We do not claim that SPOT is state of the art.**

***Index Terms*—causal discovery, latent confounding, ADMG, PAG, skeleton posterior, SPOT, ABIC, PetShop**

I. Introduction

Microservice latency data often miss services that carry no usable metric stream, so some causes are hidden and a plain DAG is the wrong model. Latent-aware learners represent observed services with an acyclic directed mixed graph (ADMG): directed edges for measured propagation, bidirected edges for dependence caused by unobserved common causes. The differentiable ADMG learner ABIC [3] fits such graphs but searches over all d^2 candidate edges, which is slow as d grows. SPOT [4] trains a skeleton posterior offline on simulated graphs and uses it to mask and prune the search.

This project is an independent evaluation of SPOT against its ABIC backbone, on simulated ancestral ADMGs and on the PetShop benchmark [5]. We ask where posterior guidance helps, where it does not, and what limits it, and we report negative results as they came. Contributions: (1) a re-implementation of ABIC and SPOT with a shared optimizer so the posterior is the only difference; (2) a paired, pre-declared statistical protocol with its limits stated; (3) results on accuracy, speed, probability quality, held-out prediction, classical baselines, root-cause ranking and scalability; (4) a repositioned claim that is narrower than the original.

II. Problem Formulation

Given n observations X of d observed services, recover the structure of the ADMG over them. The target is scored as the PAG of the true graph: skeleton, arrowheads and tails, plus PAG structural Hamming distance (SHD). A second problem is predictive: given a fitted graph, infer the distribution of one variable from others. A PGM is appropriate because the dependence structure among services, including confounding by unobserved ones, is the object of interest, and the fitted model also answers conditional queries.

Hypotheses fixed in advance: **H1** SPOT improves skeleton accuracy over ABIC on simulated ADMGs and the gain shrinks with n; **H2** the PetShop gain is smaller than the synthetic gain; **H3** the speed-up comes mainly from the hard floor; **H4** the posterior is calibrated on simulated data and over-confident on PetShop; **H5** a fitted ADMG predicts held-out data better than no structure, and accurate structure helps root-cause analysis.

III. Background

ADMGs and PAGs. An ancestral ADMG has directed and bidirected edges, with no bidirected edge between a node and one of its ancestors. Its maximal ancestral graph (MAG) has one edge for every inducing path, and its Markov equivalence class is summarized by a PAG; skeleton, arrowheads and tails of the PAG are what we score [6], [7].

Linear-Gaussian ADMG. $X = XW + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \Omega)$; the support of W gives directed edges and the off-diagonal support of Ω gives bidirected ones, so

$$\Sigma = (I - W)^{-\top}\, \Omega\, (I - W)^{-1}. \tag{1}$$

Differentiable discovery. NOTEARS [2] replaces the combinatorial acyclicity constraint with a smooth penalty; ABIC [3] extends it to ADMGs with an ancestrality constraint and covariance updates, solved with an augmented Lagrangian.

IV. Related Work

Constraint-based: FCI and RFCI [8], [9] recover PAGs from conditional-independence tests; they are fast and, on our

linear-Gaussian synthetic graphs, more accurate than the score-based methods. *Score-based and differentiable:* NOTEARS [2] and ABIC [3]. *Skeleton-guided:* SPOT [4], the method under test. *Benchmark:* PetShop [5], a microservice dataset with a call graph and injected issues, used for structure learning and root-cause analysis.

V. SOTA Positioning

Anchor. SPOT (Ma et al., KDD 2024; code in Microsoft's `reliableai` repository) is our research-lab anchor. It claims scalable differentiable causal discovery under latent confounders by guiding an ABIC-style optimizer with a learned skeleton posterior.

What we did. We re-implemented the optimizer and the posterior cascade, kept the optimizer, objective, restarts, thresholds and seeds identical for ABIC and SPOT, and compared them on simulated ADMGs and PetShop with FCI/RFCI as classical baselines. Our posterior is a scikit-learn histogram gradient-boosting cascade, not the released xgboost one, so its floor was calibrated separately (6×10^{-3} instead of 10^{-3}).

Revisited after the results. Table I gives the repositioned claims. The defensible contribution is an independent, application-specific evaluation of posterior-guided optimization on a real microservice benchmark: where it helps (speed, sparse synthetic graphs), where it does not (classical baselines, root-cause ranking, a posterior that transfers imperfectly) and what limits it (posterior quality). This differs from a claim of general superiority; we consider the narrower claim the one the evidence supports.

VI. Graphical Representation

Variables are the observed services' latency series. The model is the linear-Gaussian ADMG above: W holds directed edges (callee latency to caller latency, so the causal direction reverses the call direction) and Ω holds bidirected edges for latent common causes. Unobserved services are not modeled explicitly; their effect enters as bidirected edges after latent projection. The learned graph implies the independences of its m-separation relations and is converted to a MAG and a PAG for comparison. PetShop services without a usable metric stream make latent variables a property of the application, which a DAG cannot express. Representation size is reported as the number of free pairs after pruning (118 for SPOT-cal and 203 for SPOT at 10^{-3} on PetShop, against all pairs for ABIC) and as the estimated directed and bidirected edge counts.

VII. Learning Method

Parameters and structure (ABIC). Minimize a least-squares data term with a smooth ℓ_0 penalty under acyclicity and ancestrality constraints, using an augmented Lagrangian and L-BFGS-B; error-covariance pseudo-variables are updated with RICF. Structure and parameters are learned jointly.

Skeleton posterior (SPOT). A three-stage gradient-boosting cascade estimates $p_{ij} = P(i \sim j \mid X)$ from correlation-based features: stage 0 uses Markov-blanket features, stage 1 adds order-1 conditional-independence summaries for pairs passing a first threshold (0.05), stage 2 adds order-2 summaries for pairs passing a second (0.10); out-of-fold predictions feed the next stage. It is trained offline on simulated ancestral ADMGs (d 25–40, n 300–2000, AR(1) 0/0.3/0.6) with labels from the maximal ancestral projection of the true graph.

Guided optimization. The gradient for a pair is masked according to its posterior and the direction of the proposed update; an optional hard floor pins pairs with p below the floor to zero. The floor was calibrated on held-out simulated graphs (seed 777, d 25–40) as the largest floor in a fixed grid pinning at most 1% of true edges, giving 6×10^{-3} (released code: 10^{-3} for its xgboost cascade); benchmark graphs use other seeds. Pruning improbable pairs shrinks the parameter set the optimizer searches, which is where the time goes.

VIII. Inference Method

Problem. Given a fitted ADMG (W, Ω), predict a target service's latency from observed evidence variables with a calibrated predictive distribution.

Algorithm. The fitted model implies a joint Gaussian with covariance Σ above. The predictive distribution of target s given evidence e is the Gaussian conditional with mean $\Sigma_{se}\Sigma_{ee}^{-1}x_e$ and variance $\Sigma_{ss} - \Sigma_{se}\Sigma_{ee}^{-1}\Sigma_{es}$. This is exact for the fitted model; the cost is one linear solve of size $|e|$ per query, so the model fit, not the query, dominates runtime.

Evaluation (E8). Rolling time-ordered splits; query A conditions on the target's true ancestors, query B on all other variables. Models: true ADMG, ABIC ADMG, SPOT-cal ADMG, DAG fits, saturated Gaussian, marginal. KPIs: held-out NLL, 90% interval coverage, variance ratio, PIT KS statistic. Root-cause ranking (E12) is a second use of the graph. Exact-vs-approximate inference was planned and not run.

IX. Experimental Setup

Data. *Synthetic:* random ancestral ADMGs with linear-Gaussian data (directed weights $\pm U[0.5, 2.0]$, error covariances $\pm U[0.4, 0.7]$, diagonally dominant), $d = 25$, $n \in \{200, 500, 1000, 2000\}$ (E1, 10 graphs per n), plus E2 robustness cells at $n = 1000$ (density, SNR, AR(1) 0/0.3/0.6, 5% and 20% missing data). *PetShop:* four scenarios (low_traffic, high_traffic, temporal_traffic1, temporal_traffic2) with about 29–35 observed and 9–13 latent services; normal-operation latency averages for structure learning, injected-issue runs for root-cause ranking; one dataset per scenario; Gaussian-rank preprocessing (raw-centered in E13).

Runs. E3: 4 scenarios $\times$ 5 seeds $\times$ {ABIC, SPOT, SPOT-cal}. E4: ablations, 3 seeds. E5: 20 moving-block bootstrap resamples for each of 3 scenarios. E6: equal budgets of 60 s and 120 s on a quiet machine. E7: posterior quality. E8: inference validation, 20 units per model. E9: FCI/RFCI (Fisher-z, $\alpha \in \{0.01, 0.05\}$, Tetrad `causal-cmd`). E10: directed-only learner. E12: root-cause ranking, 36 cases. E13: preprocessing. Scalability: $d \in \{25, 50, 100\}$, $n = 1000$, 600 s per run, 3–4 graphs per d.

TABLE I
POSITIONING AGAINST THE SPOT CLAIMS, REVISITED AFTER THE EXPERIMENTS.

Question	Our evidence	Position
Beats its ABIC backbone?	Synthetic skeleton F1 +0.067 (supported); PetShop +0.029 pooled (small, not detectable per scenario)	Yes, modestly; more on synthetic than real data
Faster?	About 8× lower mean time on PetShop; converged in 11/12 vs 1/12 runs at an equal 60 s budget	Yes; the robust result
Scales?	$d = 25$ to 100: higher skeleton F1 than scaled-start ABIC on every seed (3–4 seeds); $d = 200$ not completed	Encouraging, partial
Better than classical methods?	FCI/RFCI reach F1 0.87–0.89 vs 0.756 on synthetic graphs, and are comparable on PetShop	No; we do not claim state of the art
Learned structure helps downstream?	Learned graphs did not improve root-cause ranking; they improve held-out prediction over no structure	Not shown for RCA

Scoring. The learned ADMG is converted to a MAG and a PAG and compared with the true PAG (PetShop: reversed call graph, Verma–Pearl latent projection onto observed services, maximal ancestral projection, PAG).

Reproducibility. Python 3.13, scikit-learn 1.9.1, Java 17+ with `causal-cmd` 1.3.0. Everything runs from `Code/src` with `bash run_all.sh main <workers>` then `bash run_all.sh quiet <workers>`; seeds are stored in each result row. Main-pass rows predate the environment stamps; E6, E8 and scalability rows carry code hash, library versions, CPU model and load average.

X. BASELINES

ABIC [3]: the unguided differentiable ADMG learner and our optimizer backbone. **ABIC scaled start** (`abic_s`): same optimizer started at $0.5\sqrt{25/d}$ of the default scale so the acyclicity term does not overflow at large d; used only in the scalability sweep. **SPOT-cal**: ABIC plus the skeleton posterior (mask and floor 6×10^{-3}); **SPOT**: the released floor 10^{-3}. **FCI/RFCI**: classical constraint-based methods. **Ablations**: floor only, mask only, random posterior, oracle posterior, directed-only learner. **Inference baselines**: saturated Gaussian, DAG fit, marginal; root-cause baselines: anomaly size and correlation. ABIC and SPOT share optimizer, objective, restarts, thresholds and seeds.

XI. KPIS

Structure: skeleton, arrowhead and tail F1; PAG SHD (always beside F1); directed and bidirected edge counts. **Cost:** optimizer wall time, convergence within a fixed budget, peak memory, posterior time, free pairs. **Probability quality:** log-loss (headline), Brier score, ECE, AUROC, AUPRC. **Inference:** held-out NLL, 90% coverage, variance ratio, PIT KS. **Application:** top-1/top-3 recall and mean reciprocal rank (MRR). **Statistics:** mean ± SD, paired differences with 95% CI, paired t and Wilcoxon tests, moving-block bootstrap. AIC/BIC and exact-vs-approximate inference error were planned and not computed.

XII. RESULTS

All numbers come from the aggregated result tables (ids in brackets). "Supported" follows the fixed decision rule: paired t-test $p < 0.05$, Wilcoxon $p < 0.05$ and a 95% CI excluding 0; otherwise "not distinguishable from noise". Main-pass times were measured with 11 jobs running at once, so only ratios inside the main pass are meaningful; E6 and E8 give the clean timings.

TABLE II
E1: SKELETON F1 AND PAG SHD, 10 GRAPHS PER n [T1A]. †: NOT SUPPORTED.

n	F1 ABIC	F1 SPOT-cal	Δ	SHD ABIC	SHD SPOT-cal
200	0.619	0.707	+0.088	47.7	40.9
500	0.652	0.696	+0.045†	42.4	36.8
1000	0.676	0.756	+0.079	38.5	33.9
2000	0.674	0.731	+0.058	37.9	31.0

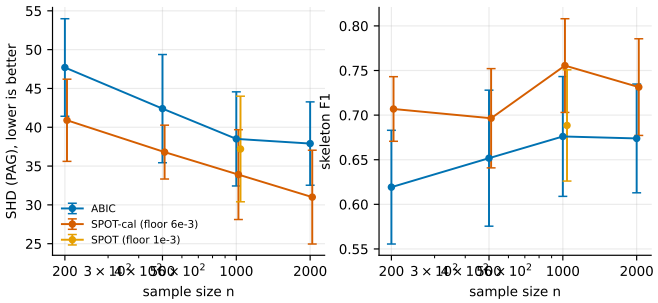


Fig. 1. E1: PAG SHD (left) and skeleton F1 (right) against sample size, mean with error bars over 10 graphs.

A. Synthetic graphs (E1, E2; $d = 25$)

Pooled over 40 (graph, n) units, SPOT-cal improves skeleton F1 by $+0.067$ (95% CI $[+0.044, +0.090]$; 31 wins, 9 losses) [T3]. The gain is supported at $n = 200, 1000, 2000$ but not at $n = 500$ (Table II, Fig. 1). We found no evidence that the gain shrinks with n (slope -0.0057 per doubling, CI $[-0.025, +0.013]$, interaction $p = 0.84$) [T4]. Time on synthetic data is only modestly lower for SPOT-cal ($n = 1000$: 127 s vs 181 s); the large speed-up is a PetShop result.

The calibrated floor matters: at $n = 1000$ SPOT with the released floor 10^{-3} gains $+0.012$ $[-0.024, +0.048]$ (not supported), against $+0.079$ for 6×10^{-3} (supported). In the robustness study (E2, $n = 1000$, 79 of 80 jobs run) SPOT-cal has higher mean skeleton F1 in all 10 conditions, supported in

TABLE III
E3: PETSHOP SKELETON F1, FIVE OPTIMIZER SEEDS PER SCENARIO [T2].

Scenario	ABIC	SPOT-cal
low traffic	0.491	0.532
high traffic	0.498	0.501
temporal traffic 1	0.538	0.561
temporal traffic 2	0.526	0.574
mean of the four	0.513	0.542

6 of 10 (density 1, 2, 3; SNR 1, 3; AR 0.3), and not supported at SNR 10, AR 0.6 and 5% or 20% missing data (Fig. 2). The gain is largest for the sparsest graphs (density 1: $+0.093$) and shrinks as density rises ($+0.049$, then $+0.030$); accuracy of both methods falls sharply with density (ABIC F1 0.74 to 0.41) [T1b].

B. PetShop accuracy (E3, E5)

Pooled over 20 (scenario, seed) units, SPOT-cal improves skeleton F1 by $+0.029$ (CI $[+0.014, +0.045]$) and lowers PAG SHD by 3.9 (CI $[-5.4, -2.4]$) [T3]. SPOT-cal wins skeleton F1 in all four scenarios on the mean (Table III), but no single scenario passes the decision rule: five seeds cannot reach Wilcoxon $p < 0.05$ (minimum about 0.0625), a property of the design, not evidence of no effect. Arrowhead ($+0.024$) and tail ($+0.031$) differences are not significant, and tail F1 is near zero for every method (ABIC 0.040, SPOT-cal 0.071), so we make no orientation claim. Both methods return 20–40 bidirected edges per graph against 31–38 true adjacencies, which keeps PetShop SHD high (about 40–60).

Under data resampling (E5, moving-block bootstrap, three scenarios, 60 pooled units; Fig. 3) the gain is positive in 40 of 58 non-tied resamples with mean $+0.022$, and its interval spans zero, so we do not call the PetShop gain significant under resampling. The E5 t and Wilcoxon p-values (about 0.001 and 0.002) are not used: the resamples overlap and the interval is the spread across resamples, not a confidence interval for the mean. The PetShop gain is smaller than the synthetic gain (difference -0.057, CI $[-0.097, -0.017]$, Welch $p = 0.009$; H2 supported); the out-of-distribution posterior (AUROC 0.879 vs 0.990) is a plausible reason, not a proven one.

C. Speed and equal budget (E3, E4, E6)

Over the 20 units, mean optimizer time falls from 376 s (ABIC) to 47 s (SPOT-cal), about $8\times$ lower; the paired difference is -329 s (CI $[-454, -204]$) with SPOT-cal faster in 19 of 20 units [T1, T3]. These seconds were measured under load. On a quiet machine (E6, 12 runs per cell, 43 of 48 rows timing-clean) ABIC converged in 1 of 12 runs at a 60 s budget and SPOT-cal in 11 of 12; at 120 s SPOT-cal converged in 12 of 12 and ABIC in 6 of 12 (7 of 12 hit the budget; one run is flagged both converged and timed out) (Fig. 4). At 60 s the skeleton-F1 difference is $+0.035$ $[0.001, 0.069]$ ($p_t = 0.045$, $p_W = 0.064$: not supported) and the SHD difference -2.3 (supported); at 120 s F1 is $+0.016$ (not significant) and SHD -3.25 (supported). At an equal budget SPOT-cal reaches a lower SHD; its skeleton F1 is not significantly different [T2d].

TABLE IV
E4 ABLATION [T3_ABLATION]: TIME RELATIVE TO ABIC, SKELETON F1 (MEAN OF 12 UNITS).

Variant	Time ratio	Skeleton F1
ABIC	1.00	0.511
floor only	0.23	0.515
mask only	0.40	0.527
SPOT-cal (mask + 6×10^{-3})	0.15	0.532
SPOT (10^{-3})	–	0.530
random posterior	–	0.196
oracle posterior	–	0.813

TABLE V
E7: SKELETON-POSTERIOR QUALITY. SIMULATED: 31,888 PAIRS, 2,465 TRUE EDGES.

	Simulated	PetShop (4 scen.)
log-loss (headline)	0.052	0.26–0.32
Brier	0.0136	0.053–0.067
ECE	0.0015	0.052–0.069
AUROC	0.990	0.84–0.92
AUPRC	0.943	0.53–0.59

Ablation (E4, 12 matched units; Table IV, Fig. 5). Both components speed up the optimizer, and the floor alone recovers most of the saving (median share 0.95), so H3 is partly supported. Only the random (-0.316) and oracle ($+0.302$) variants differ from ABIC on skeleton F1; SPOT-cal vs ABIC on these 12 units is $+0.020$ $[-0.003, +0.043]$ (not supported), while SHD -3.0 $[-4.7, -1.3]$ is. Free pairs after pruning: 118 (SPOT-cal), 203 (SPOT at 10^{-3}).

D. Posterior quality (E7)

Table V and Fig. 6: the posterior is calibrated on simulated graphs and over-confident on PetShop (over-confidence $+0.048$ to $+0.051$, all CIs above 0) [T7]. Isotonic and Platt recalibration do not reduce PetShop ECE (0.050–0.073) and isotonic worsens log-loss (low traffic: 0.434 vs 0.321). PetShop intervals treat edge pairs as independent and are too narrow; with one graph per scenario they show no replication. Brier intervals were not computed (that file predates the Brier bootstrap).

E. Held-out prediction (E8)

Query B predicts the sink from all other variables; 20 units per model; held-out NLL in nats (lower is better) [T5], Table VI. A fitted ADMG predicts better than no structure: $+0.747$ nats over the marginal for ABIC $[0.585, 0.909]$, $+0.745$ for SPOT-cal $[0.598, 0.893]$ (truth $+0.792$); ADMGs beat DAGs by $+0.40$ (ABIC) and $+0.37$ (SPOT-cal), both supported. Non-inferiority to the saturated Gaussian holds only for the true graph (CI lower bound -0.031 against a margin of -0.05); for learned graphs it is inconclusive (-0.083, -0.081). Coverage of the 90% interval is 0.925 and the variance ratio 1.43–1.46 (1 = calibrated): predictive variance

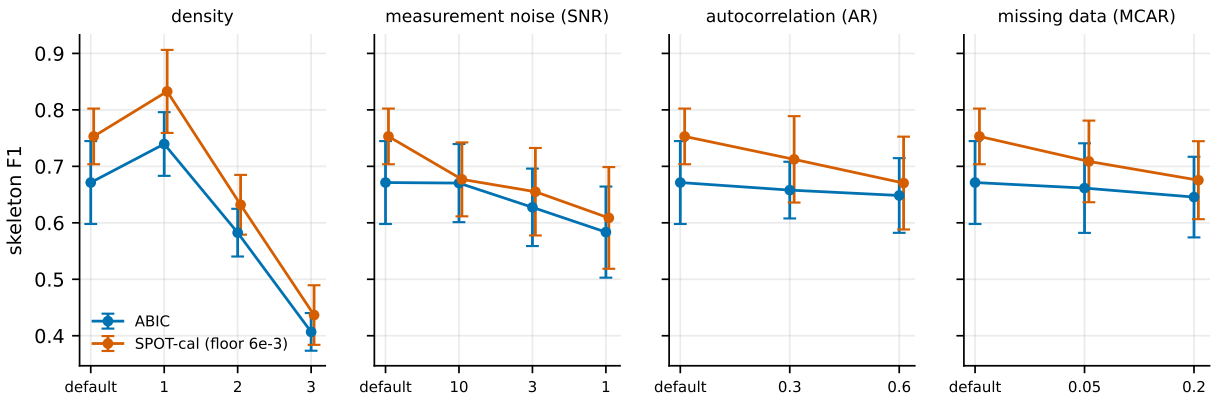


Fig. 2. E2: robustness of ABIC and SPOT-cal at $n = 1000$ to graph density, measurement noise, autocorrelation and missing data.

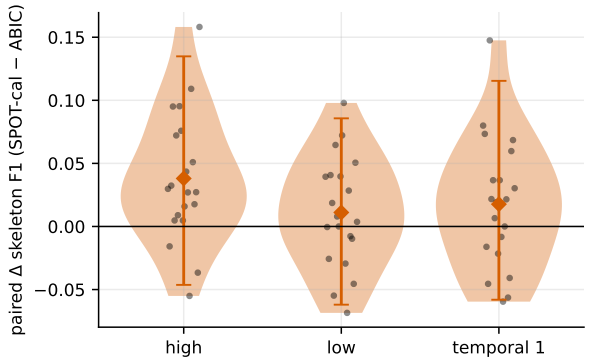


Fig. 3. E5: paired skeleton-F1 differences (SPOT-cal minus ABIC) across moving-block bootstrap resamples.

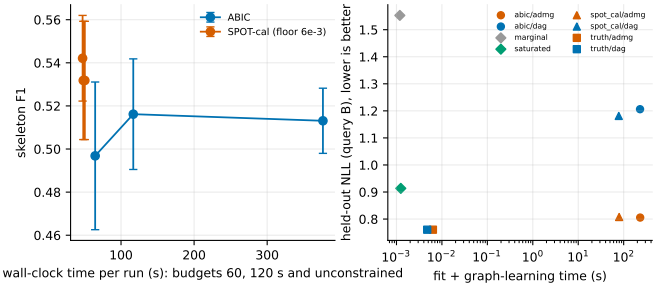


Fig. 4. Left: skeleton F1 against wall-clock time per run (E6 budgets 60 s, 120 s and unconstrained). Right: held-out NLL (query B) against fit and graph-learning time (E8).

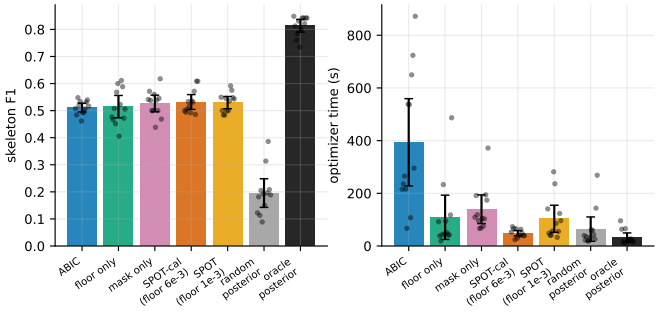


Fig. 5. E4: ablation of the mask, the floor, a random posterior and an oracle posterior on PetShop.

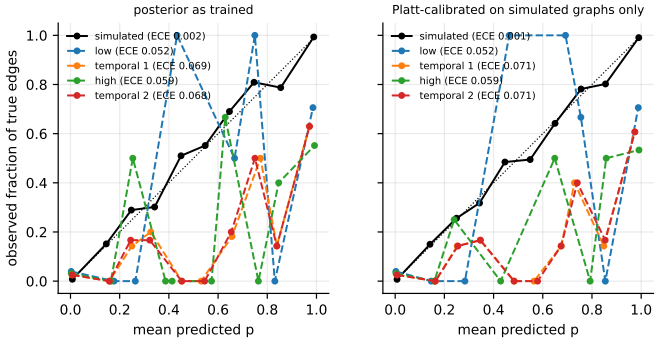


Fig. 6. E7: reliability diagrams of the skeleton posterior on simulated graphs and PetShop.

is somewhat under-estimated. Query A is inconclusive (NLL SD about 2, no significant contrast).

F. Classical baselines (E9)

Table VII. FCI and RFCI beat SPOT-cal on skeleton F1 on all 10 graphs at every n and α (differences $+0.08$ to $+0.18$) [T3]. On PetShop all methods sit near 0.5: FCI 0.497/0.500 and RFCI 0.518/0.508 ($\alpha = 0.01/0.05$) against SPOT-cal 0.542 and ABIC 0.513; with four scenario rows this is descriptive only.

G. Downstream use and sensitivity (E10, E12, E13)

Root-cause ranking (E12, 36 cases, 26 in the test split). The true ADMG improves MRR over an anomaly-size baseline by $+0.068$ $[0.001, 0.149]$ (0.68 vs 0.61). Learned graphs do not: ABIC -0.017, SPOT-cal -0.044 (both not significant), and on the test split the SPOT-cal ADMG ranks worse than the correlation baseline (-0.148 $[-0.297, -0.009]$). The true graph beats the ABIC graph by $+0.084$ and the SPOT-cal graph by $+0.112$ (both supported) [T6b]. Structure helps only when it is accurate. **Directed-only learner (E10).** Drop-

TABLE VI
E8: HELD-OUT NLL, QUERY B.

Model	NLL
true ADMG	0.761
ABIC ADMG	0.806
SPOT-cal ADMG	0.808
saturated Gaussian	0.913
ABIC DAG	1.206
SPOT-cal DAG	1.181
marginal (no structure)	1.553

TABLE VII
E9: SYNTHETIC GRAPHS, $n = 1000$.

Method	Skeleton F1	PAG SHD	Time
FCI ($\alpha = 0.01$)	0.869	17.9	≈3.3 s
RFCI ($\alpha = 0.05$)	0.885	18.8	≈3.3 s
SPOT-cal	0.756	33.9	127 s
ABIC	0.676	38.5	181 s

ping bidirected edges lowers skeleton F1 for ABIC (-0.054 $[-0.091, -0.017]$) and arrowhead F1 for both methods (ABIC -0.252, SPOT-cal -0.271); its lower SHD is an artifact of dropping edges. **Preprocessing (E13).** Gaussian-rank: ABIC 0.513, SPOT-cal 0.542 (20 units); raw-centered: 0.217 and 0.303 (12 units); the transform is essential for both methods, and the blocks differ in units, so this is descriptive [T2f].

H. Scalability (partial)

Synthetic graphs, $n = 1000$, 600 s per run, 3–4 graphs per d (Table VIII). `abic` uses the default dense start, `abic_s` a start scaled by $\sqrt{25/d}$ (identical at $d = 25$). At $d = 100$ the two working variants hit the budget, so those rows show quality reached in fixed time, not converged solutions. Results are descriptive; no p-values are attached. With its default start, plain ABIC diverges at $d = 100$ on 3 of 3 graphs and returns an empty graph after about 15 s. Against scaled-start ABIC, SPOT-cal's skeleton F1 is higher on every seed at $d = 25$ ($+0.087$, 4/4), 50 ($+0.067$, 3/3) and 100 ($+0.044$, 3/3), and its mean SHD is lower by 6, 12 and 49. The empty graph at $d = 100$ has a lower SHD than `abic_s` (131 vs 157), so SHD must be read beside F1. The posterior step costs 0.4 s at $d = 25$ and 5.2 s at $d = 100$; peak memory is about 200–370 MB. $d = 200$ **was not completed** (the workers hung and were stopped), and the wide-posterior variant did not run, so we make no claim about $d = 200$ or a posterior trained on larger graphs.

XIII. STATISTICAL ANALYSIS

Every comparison is paired (same graph or scenario and seed). A difference is supported only if the paired t-test $p < 0.05$, the Wilcoxon signed-rank $p < 0.05$ and the 95% CI of the paired difference excludes 0; the rule was fixed before the results. Limits: (1) *PetShop seeds measure optimizer variance only*; each scenario is one fixed series, and five seeds cannot reach Wilcoxon $p < 0.05$, so we rely on the pooled 20 units. (2) *E5 is not a confidence interval for the mean gain*: resamples overlap and its interval is the spread across resamples. (3) *Pseudo-replication:* E1 pools 10 graphs $\times$ 4 sample sizes, so per-n tests are cleaner; PetShop posterior intervals treat edge pairs as independent. (4) *Many comparisons:* the paired-test table holds 331 comparisons; a Holm-adjusted column exists but verdicts do not use it, so single p-values are not headlines. (5) *SHD can mislead:* a sparse or empty graph can have low SHD (random-posterior variant 42.9 vs ABIC 48.4 with F1 0.196), so F1 is always shown beside SHD. (6) Scalability has 3–4 graphs per d and FCI/RFCI on PetShop has 4 scenario rows; both are descriptive.

TABLE VIII
SCALABILITY: MEAN OVER 3–4 GRAPHS, 600 S BUDGET (PARTIAL).

d	Variant	F1	SHD	Budget hit	Time (s)
25	ABIC	0.677	42.8	0/4	180
25	ABIC scaled	0.677	42.8	0/4	167
25	SPOT-cal	0.763	36.5	0/4	121
50	ABIC	0.666	79.0	1/3	515
50	ABIC scaled	0.692	76.3	0/3	423
50	SPOT-cal	0.759	64.3	1/3	567
100	ABIC	0.000	131.3	0/3	17
100	ABIC scaled	0.708	157.0	3/3	675
100	SPOT-cal	0.753	107.7	3/3	663

XIV. DISCUSSION

Speed. The optimizer time ratio against ABIC is 0.23 with the floor alone, 0.40 with the mask alone and 0.15 with both. Pinning improbable pairs removes parameters from the search, which is consistent with the saving. The quiet-machine E6 run confirms it independently of load. The ablation cannot attribute accuracy to either component: no variant differs from ABIC on skeleton F1.

Accuracy. The synthetic gain ($+0.067$) is supported; the PetShop gain ($+0.029$) is smaller and not detectable per scenario. The gain depends on a floor calibrated for our own posterior. A plausible reason for the smaller PetShop gain is the posterior's transfer: AUROC falls from 0.990 to 0.84–0.92 and the posterior becomes over-confident (ECE 0.0015 vs 0.052–0.069); we did not test that mechanism directly. The oracle posterior (F1 0.813 vs 0.542) shows the optimizer is not the ceiling; the posterior is.

Where it fails. Both score-based methods return far more bidirected edges than the truth (20–40 against 0–1), tail F1 is near zero, and learned graphs do not help root-cause ranking. FCI/RFCI are more accurate on our synthetic graphs (F1 0.87–0.89 vs 0.756) and about $40\times$ faster; we did not isolate why, but the data are linear-Gaussian, the setting the Fisher-z test is designed for. With its default dense start, our ABIC implementation diverges at $d = 100$ because its acyclicity term overflows; a scaled start fixes it. This is a start-scale issue of our implementation, not a general statement about ABIC.

Inference. A fitted ADMG predicts the held-out sink better than no structure ($+0.75$ nats) and better than a DAG ($+0.37$

to $+0.40$), but only the true graph is non-inferior to a saturated Gaussian; predictive variance is somewhat under-estimated.

XV. Limitations

Data scope: one dataset per PetShop scenario, so "generalizes to real systems" is not supported; synthetic experiments use $d = 25$ (plus the scalability sweep). **Model:** linear-Gaussian with i.i.d. rows while latency is autocorrelated and skewed; Gaussian-rank preprocessing is essential. Ground truth is the inverted call graph with latent projection, not every statistical dependency. **Floor:** the calibrated floor drives the gain; it was chosen on simulated held-out graphs and never on PetShop. **Missing or partial:** E2 is 79/80 jobs; E5 has 3 of 4 scenarios; the edge-level error analysis (E11) was skipped; scalability stops at $d = 100$ with 3–4 seeds; the wide-posterior retraining, the posterior-by-d study, exact-vs-approximate inference, structure-to-inference error and a PetShop prediction study were planned for M4 and not run. **Provenance:** main-pass rows carry no code-hash or library stamps because they were run before this feature was added to the code; E6, E8 and scalability rows do. The main pass used the scikit-learn 1.9.1 posterior and the E1/E2 graphs are disjoint from the posterior's training graphs. **Timing:** main-pass times were measured with 11 jobs at once; quote ratios, and E6/E8 for clean timings. Both methods overproduce bidirected edges, limiting arrowhead and tail interpretation.

XVI. Conclusion and Future Work

Posterior-guided optimization makes ABIC-style latent causal discovery much faster on PetShop (about $8\times$ lower mean time; 11/12 vs 1/12 converged at 60 s) with a small accuracy gain ($+0.029$ skeleton F1 pooled) and a larger gain on simulated graphs ($+0.067$). The benefit relies on a calibrated floor and on a posterior that transfers imperfectly to real data. FCI/RFCI are more accurate on our synthetic graphs, and learned graphs did not improve root-cause ranking. Next: (1) retrain the posterior on graphs up to $d = 200$ and re-test at $d = 200$; (2) close the gap to the oracle posterior (0.81 vs 0.54) with PetShop-like training graphs; (3) repeat the comparison on more than one dataset per scenario; (4) test a classical-skeleton prior (for example from FCI) in place of the learned posterior; (5) measure exact vs approximate inference cost and structure-to-inference error.

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